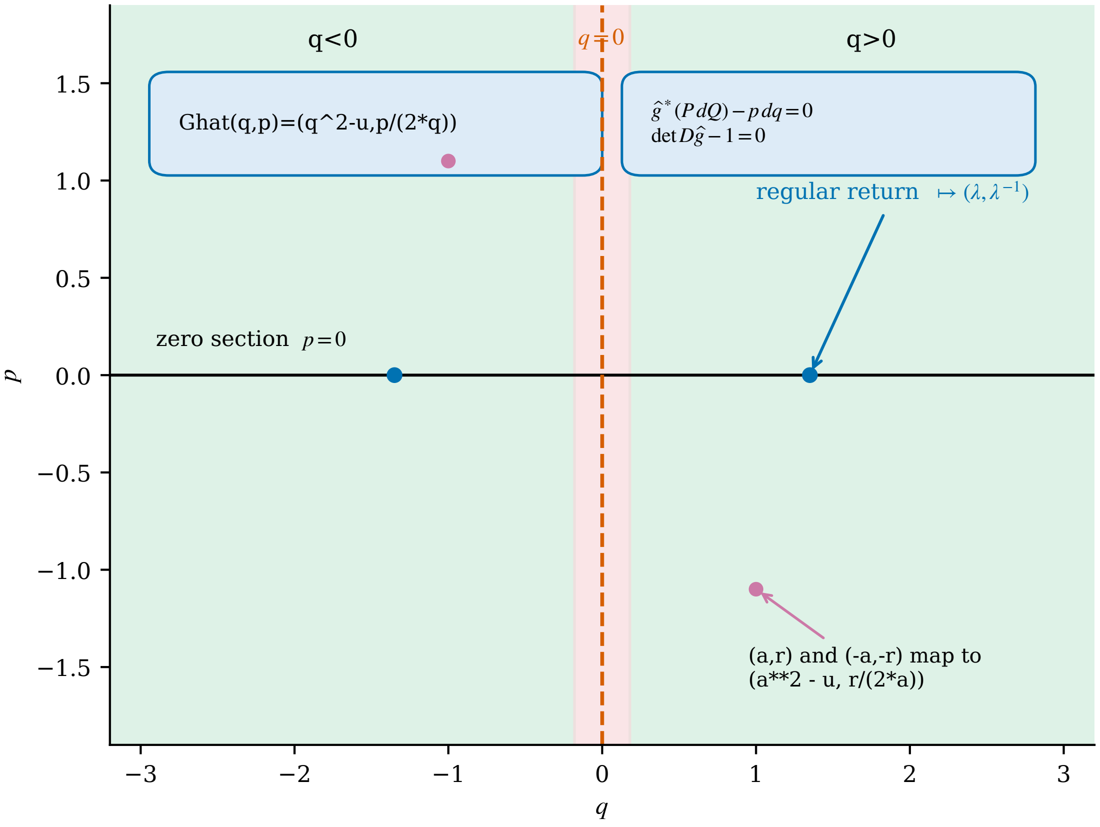
(a)



(b)

period	λ expression	check
1	`2*q0`	PASS
2	`4*q0*q1`	PASS
3	`8*q0*q1*q2`	PASS
4	`16*q0*q1*q2*q3`	PASS

- undefined at q=0
- the two regular branches have overlapping images
- noncompact phase space
- not a global symplectomorphism
- critical cycles with zero multiplier are outside the reciprocal lift

critical denominator 2*q becomes 0 at q=0; the regular branch contains (q,p)=(t,0) for every t>0 and t tends to infinity